CSMAP – 512 Research Methodology Practical Question Bank

Case Study: "An educational institution wants to study whether the amount of time spent by postgraduate students on social media affects their academic performance."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Social media usage – hours per day
   2. Frequency of social media use
   3. Purpose of social media use
   4. Study time
   5. Academic Marks (in %)

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Academic Marks (%) column. [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Solution

Q.1

a) Frame a suitable research title. [2 Marks]

Title: A Study on the Effect of Social Media Usage on the Academic Performance of Postgraduate Students.

b) State the research problem. [2 Marks]

The research problem is to investigate whether the amount of time postgraduate students spend on social media affects their academic performance. The study will examine the relationship between social media usage, study time, and academic marks.

c) Write any three research objectives. [3 Marks]

1. To determine the average number of hours postgraduate students spend on social media per day.

2. To examine the relationship between social media usage and students' academic performance.

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3. To study the relationship between students' study time and their academic marks.

d) Select a suitable research design/approach and justify your selection. [3 Marks]

Research Design: Descriptive and correlational research design.

Research Approach: Quantitative research approach.

Justification:

1. The descriptive design helps collect information about students' social media usage, study time, and academic marks.

2. The correlational design helps identify whether a relationship exists between social media usage and academic performance.

3. The quantitative approach uses numerical data and statistical techniques such as mean, median, mode, and correlation to analyse the collected information.

Q.2

a) Identify the population, sample and suitable sampling technique. [2 Marks]

Population: All postgraduate students studying at the selected educational institution.

Sample: 30 postgraduate students selected from the institution.

Sampling Technique: Simple random sampling, in which each postgraduate student has an equal chance of being selected.

b) Prepare a suitable dataset and save it in CSV format. [5 Marks]

The following sample dataset contains 10 postgraduate students and the five required variables.Create this csv in excel.

Student ID	Social Media Hours/Day	Frequency	Purpose	Study Time (Hours/Day)	Academic Marks (%)
S01	2	Occasional	Communication	4	78
S02	5	Very Frequent	Entertainment	2	65
S03	3	Frequent	Education	4	82
S04	6	Very Frequent	Entertainment	1	70
S05	4	Frequent	Education	3	75
S06	5	Very Frequent	Entertainment	2	65
S07	1	Occasional	Education	5	90
S08	4	Frequent	Communication	3	72
S09	2	Occasional	Education	4	80
S10	6	Very Frequent	Entertainment	1	68

c) Calculate the Mean, Median and Mode of Academic Marks (%). [3 Marks]

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Academic Marks: 78, 65, 82, 70, 75, 65, 90, 72, 80, 68.

1. Mean

Mean = Sum of all observations / Total number of observations

Mean = 745 / 10

Mean = 74.5%

2. Median

Arrange the marks in ascending order:

65, 65, 68, 70, 72, 75, 78, 80, 82, 90

Since the number of observations is even (N = 10), the median is the average of the 5th and 6th observations.

Median = (72 + 75) / 2

Median = 73.5%

3. Mode

The mode is the value that occurs most frequently.

The value 65 occurs twice, while all other values occur only once.

Mode = 65%

Result: Mean = 74.5%, Median = 73.5%, and Mode = 65%.

d) Using Python, read the CSV dataset and represent the data using a suitable graph/chart. [5 Marks]

Aim: To read the CSV dataset using Python and display a scatter plot showing the relationship between social media usage and academic marks.

Python Program:

```
import pandas as pd
import matplotlib.pyplot as plt
data = pd.read_csv("postgraduate_social_media_dataset.csv")

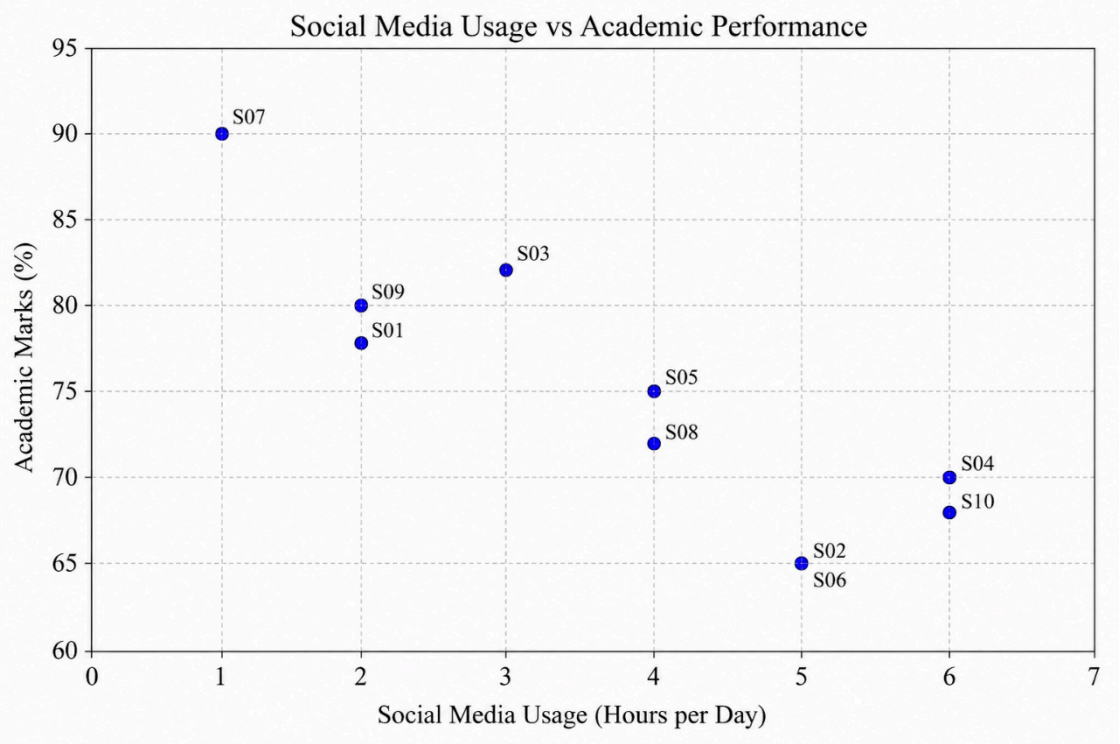
print(data)
plt.scatter(data["Social_Media_Hours_Per_Day"],
        data["Academic_Marks_Percent"])

plt.xlabel("Social Media Usage (Hours per Day)")
```

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```
plt.ylabel("Academic Marks (%)")
plt.title("Social Media Usage vs Academic Performance")

plt.grid(True)
plt.show()
```



Case Study: "An e-commerce company wants to study whether the frequency of online shopping influences customer satisfaction."Based on this research problem answer the following questions:

Q 1. a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]

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d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Online shopping frequency
2. Monthly online spending
3. Preferred product category
4. Delivery satisfaction
5. Overall customer satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Customer Satisfaction Score column. [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart. [5 Marks]

Solution

a) Frame a suitable research title. [2 Marks]

A Study on the Influence of Online Shopping Frequency on Customer Satisfaction.

b) State the research problem. [2 Marks]

The research problem is to investigate whether the frequency of online shopping influences customer satisfaction. The study examines the relationship between shopping frequency, monthly spending, preferred product category, delivery satisfaction, and overall customer satisfaction.

c) Write any three research objectives. [3 Marks]

1. To determine the online shopping frequency of customers.
2. To examine the relationship between online shopping frequency and overall customer satisfaction.
3. To study the relationship between delivery satisfaction and overall customer satisfaction.

d) Select a suitable research design/approach and justify your selection. [3 Marks]

Research design: Descriptive and correlational research design.

Approach: Quantitative research approach.

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Q.2

a) Identify the population, sample and suitable sampling technique. [2 Marks]

- Population: All customers who purchase products through the selected e-commerce company.
- Sample: 10 customers selected from the company's customer base.
- Sampling technique: Simple random sampling, in which every customer has an equal chance of being selected.

b) Prepare a suitable dataset and save it in CSV format. [5 Marks]

The following dataset contains records of 10 customers with the five required variables.

Customer ID	Shopping frequency	Monthly spending (₹)	Preferred category	Delivery satisfaction	Satisfaction score
C01	Weekly	5000	Electronics	Satisfied	5
C02	Monthly	2000	Clothing	Neutral	3
C03	Daily	9000	Groceries	Very Satisfied	5
C04	Weekly	4500	Clothing	Satisfied	4
C05	Monthly	2500	Home & Kitchen	Neutral	3
C06	Weekly	6000	Electronics	Very Satisfied	5
C07	Rarely	1000	Books	Dissatisfied	2
C08	Daily	10000	Groceries	Very Satisfied	5
C09	Monthly	3000	Clothing	Satisfied	4
C10	Weekly	5500	Home & Kitchen	Satisfied	4

The satisfaction score is measured on a scale of 1 to 5, where 1 indicates very low satisfaction and 5 indicates very high satisfaction.

CSV file: `ecommerce_customer_satisfaction_dataset.csv`

c) Using the dataset, calculate the Mean, Median and Mode of the Customer Satisfaction Score. [3 Marks]

Customer satisfaction scores: 5, 3, 5, 4, 3, 5, 2, 5, 4, 4.

1. Mean

Mean = Sum of all observations / Total number of observations

Mean = 40 / 10 = 4

Therefore, Mean = 4.

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2. Median

Arrange the scores in ascending order:

2, 3, 3, 4, 4, 4, 5, 5, 5, 5

Since the number of observations is even (N = 10),

Median = (5th observation + 6th observation) / 2

Median = (4 + 4) / 2 = 4

Therefore, Median = 4.

3. Mode

The mode is the value that occurs most frequently.

The score 5 occurs four times, which is more than any other score.

Therefore, Mode = 5.

Result: Mean = 4, Median = 4, and Mode = 5.

d) Using Python, read the CSV dataset and represent the data using a suitable graph/chart. [5 Marks]

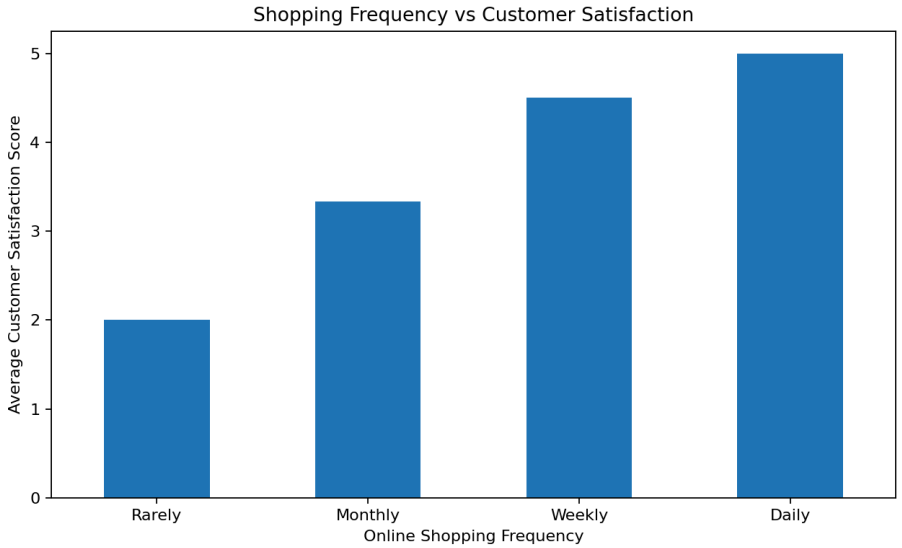
Python program:

```
import pandas as pd
import matplotlib.pyplot as plt
data = pd.read_csv("ecommerce_customer_satisfaction_dataset.csv")

result = data.groupby("Online_Shopping_Frequency")[
    "Overall_Customer_Satisfaction_Score"
].mean()

result = result.reindex(["Rarely", "Monthly", "Weekly", "Daily"])
result.plot(kind="bar")
plt.xlabel("Online Shopping Frequency")
plt.ylabel("Average Customer Satisfaction Score")
plt.title("Shopping Frequency vs Customer Satisfaction")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

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Case Study: "A company wants to study whether the number of days employees work from home is associated with their productivity.". Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Work-from-home days per week
2. Daily working hours
3. Number of meetings attended
4. Work-life balance score
5. Productivity score

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c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Productivity Score column. [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart. [5 Marks]

Case Study: "A bank wants to study the factors influencing customers' use of mobile banking services."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Mobile banking usage frequency
2. Number of transactions per month
3. Type of transaction
4. Ease-of-use rating
5. Customer satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Satisfaction Score column. [3 Marks]

d) d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart. [5 Marks]

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Case Study: "An organization wants to investigate whether employees' sleep duration is related to their work performance."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Sleep duration per night
   2. Sleep quality
   3. Working hours per day
   4. Number of breaks
   5. Work performance score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Performance Score column. [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "A community health organization wants to study whether regular physical activity is associated with self-reported well-being."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Exercise days per week
   2. Exercise duration
   3. Type of exercise
   4. Sleep duration
   5. Well-being score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Well-being Score column. [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "A city transport authority wants to study factors affecting passenger satisfaction with

Case Study: "A city transport authority wants to study factors affecting passenger satisfaction with public transportation."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Daily travel frequency
   2. Travel time
   3. Waiting time
   4. Type of public transport
   5. Passenger satisfaction score

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c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Passenger satisfaction Score column. [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "A city transport authority wants to study factors affecting passenger satisfaction with public transportation."Based on this research problem answer the following questions:

Case Study: "An agricultural research organization wants to study whether the use of modern farming technology is associated with farm productivity."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
    1. Farm size
    2. Technology used
    3. Number of technology applications
    4. Irrigation method
    5. Crop yield per acre
c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Crop yield column. [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "A training institute wants to study whether participation in online learning activities is associated with learners' test performance."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Online learning hours per week
   2. Course completion rate
   3. Number of online activities completed
   4. Self-study hours
   5. Test score (%)

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Test Score column. [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "A retail business wants to study whether customer service quality is associated with customer loyalty." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
    1. Number of visits
    2. Waiting time
    3. Service quality rating
    4. Purchase frequency
    5. Customer loyalty score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the loyalty score column. [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "A tourism department wants to study factors influencing visitor satisfaction at tourist destinations."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Number of visits
2. Length of stay
3. Accommodation type
4. Travel expenditure
5. Visitor satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Satisfaction Score column. [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "A food delivery platform wants to study whether delivery time and order frequency are associated with customer satisfaction."Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]

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d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Orders per month
   2. Average delivery time
   3. Order value
   4. Food quality rating
   5. Customer satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Satisfaction score column. [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "An organization wants to study whether participation in employee training programmes is associated with job performance." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]

b) State the research problem. [2 Marks]

c) Write any three research objectives. [3 Marks]

d) Select a suitable research design/approach and justify your selection. [3 Marks]

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Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Training hours
2. Number of training programmes attended
3. Training type
4. Job experience
5. Performance score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Performance Score column. [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "A study is proposed to examine the adoption of digital payment methods among small business owners." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]

b) State the research problem. [2 Marks]

c) Write any three research objectives. [3 Marks]

d) Select a suitable research design/approach and justify your selection. [3 Marks]

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Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Digital payment frequency
2. Monthly transaction volume
3. Preferred payment method
4. Ease-of-use rating
5. Satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Satisfaction score column . [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "A municipal authority wants to study household participation in waste segregation and recycling." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]

b) State the research problem. [2 Marks]

c) Write any three research objectives. [3 Marks]

d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Household size

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2. Waste generated per day
3. Waste segregation frequency
4. Recycling frequency
5. Waste-management awareness score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Awareness score column . [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "A health research group wants to study whether smartphone use before bedtime is associated with sleep quality." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]

b) State the research problem. [2 Marks]

c) Write any three research objectives. [3 Marks]

d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Smartphone usage before sleep
2. Duration of use
3. Bedtime
4. Sleep duration
5. Sleep quality score

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c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Sleep Quality score column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "An online marketplace wants to study whether customer reviews influence consumers' purchase decisions." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Number of reviews viewed
   2. Review rating
   3. Time spent reading reviews
   4. Product price
   5. Purchase decision score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Purchase decision score column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "A human resource research study wants to examine whether commuting time is associated with employee job satisfaction." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Daily commuting time
   2. Distance travelled
   3. Mode of transport
   4. Monthly commuting expenditure
   5. Job satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Job satisfaction score column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "A study is proposed to examine whether social media marketing activities are associated with sales performance of small businesses." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Social media marketing hours per week
   2. Number of posts per week
   3. Platform used
   4. Customer enquiries
   5. Monthly sales

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of monthly sales column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

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Case Study: "An agricultural study wants to examine factors associated with water consumption in farming." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Farm size
   2. Irrigation frequency
   3. Irrigation method
   4. Water used per day
   5. Crop yield per acre

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Crop yield column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

CSMAP – 512 Research Methodology Practical Question Bank

Case Study: "A healthcare facility wants to study factors affecting patient satisfaction with its outpatient services." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Waiting time
   2. Number of visits
   3. Consultation duration
   4. Staff behaviour rating
   5. Patient satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Patient satisfaction score column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

CSMAP – 512 Research Methodology Practical Question Bank

Case Study: "A media research organization wants to study patterns of news consumption and their relationship with users' trust in news sources." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Daily news consumption time
2. Preferred news source
3. Frequency of news consumption
4. Type of news preferred
5. Trust in news score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the trust score column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "A market research organization wants to study factors influencing consumers' intention to purchase an electric vehicle.".Based on this research problem answer the following questions:

CSMAP – 512 Research Methodology Practical Question Bank

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]
   1. Monthly travel distance
   2. Awareness of electric vehicles
   3. Charging availability
   4. Price sensitivity
   5. Purchase intention score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Purchase intention score column . [3 Marks]
d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "A public service organization wants to study whether digital literacy is associated with citizens' use of online public services." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]
b) State the research problem. [2 Marks]
c) Write any three research objectives. [3 Marks]
d) Select a suitable research design/approach and justify your selection. [3 Marks]

CSMAP – 512 Research Methodology Practical Question Bank

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Digital literacy score
2. Internet usage hours per day
3. Online service usage frequency
4. Type of online service used
5. Satisfaction score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Satisfaction score column . [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]

Case Study: "An environmental research organization wants to study whether environmental awareness is associated with household recycling behaviour." Based on this research problem answer the following questions:

Q 1.

a) Frame a suitable research title. [2 Marks]

b) State the research problem. [2 Marks]

c) Write any three research objectives. [3 Marks]

d) Select a suitable research design/approach and justify your selection. [3 Marks]

CSMAP – 512 Research Methodology Practical Question Bank

Q 2.

a) Identify the population, sample and suitable sampling technique for the proposed study. [2 Marks]

b) Prepare a suitable dataset based on the following variables/features. Create appropriate records for the variables, assign suitable values/categories, and save the dataset in CSV format. [5 Marks]

1. Environmental awareness score
2. Recycling frequency
3. Household waste generated per week
4. Waste segregation frequency
5. Recycling behaviour score

c) Using the dataset prepared in Question (b), calculate the Mean, Median, and Mode of the Recycling behaviour score column . [3 Marks]

d) Using Python, read the CSV dataset prepared in Question (b) and represent the data using a suitable graph/chart**.** [5 Marks]